having Mamata as an alliance partner would be ruinous. As a B-team of the BJP, she's working towards a Congress- and Left-*mukt* Bengal."

Political moves apart, Mamata has renewed focus on her image as a humane CM. After the railway accident in Odisha's Balasore on June 2, she visited the scene the next day. With 103 of those who died and most of the injured being from Bengal, she visited the Cuttack and Bhubaneswar hospitals where they were admitted. Back home, she rolled out a compensation package, some of which was handed out in a sombre programme in front of waiting cameras.

Mamata also made it a point to make the right noises about the wrestlers' protest in Delhi, the unrest in Manipur and arranged the evacuation of 18 students stranded in Imphal in May. "It was a conscious move to deflect attention from the ongoing corruption charges against the TMC and appeal to civil society and the educated class," says Prasanta Ray, professor emeritus of Presidency University.

THE BENGAL CM IS FOCUSING ON A HUMANE IMAGE AND HAS STARTED A NEW OUTREACH PROGRAMME

At home, Mamata has started the public outreach platform *Shorashori Mukhyomantri* (directly with the CM) in early June to hear and solve common grievances. It is a new version of her 2019 *Didi ke Bolo* (Tell Didi) initiative. Launched when there were widespread allegations of 'cut money' extortion against local-level TMC leaders, it had yielded results. *Shorashori Mukhyomantri* has been crafted along similar lines. Now, if she can give full attention to what people say. ■

GUJARAT/ MICRO EARTHQUAKES

THE FAULT LINES OF SAURASHTRA

By Jumana Shah

The topography of the arid peninsula of Saurashtra has metamorphosed into lush green in the past two decades. Thanks to good monsoons resulting in healthy agriculture and forest cover. However, little did Kathiawar expect that this boon would eventually become a bane of sorts for the region. In the past two years, many regions in the peninsula have been experiencing seismic swarms—several small earthquakes measuring less than 3.4 on the Richter scale keep recurring for an extended period of time. Despite being low-intensity, their strange recurrence has left residents unnerved.

A study by the Institute of Seismological Research (ISR), Gandhinagar, indicates these quakes do not necessarily presage a bigger one. But it underlines that earthquakes cannot be predicted despite the leap in technology. ISR director Dr Sumer Chopra explains that Gujarat and the surrounding states rest on the Deccan Traps, a large area of volcanic lava cooled into rock, but with a lot of fractures. "The region is being subjected to hydrological loading in addition to tectonic loading. Amreli, for instance, has been experiencing above-average rainfall for the past few years. The region is prone to moderate and low-magnitude earthquakes within the depth range of ~ 3 to ~ 24 km," Chopra tells INDIA TODAY.

A previous study has indicated that swarm earthquakes in Jamnagar were observed in the monsoon season, following heavy rains that drastically raised the water table by ~ 24 m, thereby increasing the pore pressure by ~ 2.0 bars. The stress changes are sufficient to trigger small earthquakes.

One can imagine that water would soothe the burning core of the earth, but apparently not. The Deccan Traps are hard, as they are about 65 million years old—compared even to the Himalayan plate, which started forming about 50 million years ago. Due to active tectonism in this part of India, the Deccan Traps are highly fractured and act as conduits for the percolation of water into these cleavages, creating additional stress.

Parts of the Amreli district alone have witnessed some 400 minor quakes, and counting, between 2021 and 2022. A peer-reviewed study by Kothyari et al. (2023) detailing these observations was recently published in the reputed *Elsevier* journal. Other regions like Talala in Gir Somnath, parts of Bhavnagar and Jamnagar districts in Saurashtra, Navsari in South Gujarat, western parts of Madhya Pradesh and regions near Nashik and Palghar in Maharashtra have been experiencing similar seismic swarms since around 2006. Parts of Saurashtra like Kalavad, Lalpur, Bhavnagar and Rajkot areas had earlier experienced swarm activity in 2001 and 2004-2005.

The fault line of major quakes runs into 60-70 km, while the faults